

# Same runtime, different altitudes

The three things are not a product menu. They are one stack, and you can move between them without a rewrite.

BUILT HERE

## LangChain

The framework. Building blocks and `create_agent` — a configurable model-and-tool harness with middleware.

WHEN YOU REACH FOR IT

Default starting point. Nearly every agent, including this one.

## LangGraph

The runtime underneath. Durable execution, state, checkpointing, streaming, human-in-the-loop.

WHEN YOU REACH FOR IT

Directly, when you need explicit state transitions, deterministic routing, or parallel branches.

## Deep Agents

An opinionated harness on top. Planning, subagents, virtual filesystem, skills.

WHEN YOU REACH FOR IT

Long-horizon work where the context outgrows the window.

**You don't pick one.** `create_agent` returns a compiled LangGraph. What I built today drops into a `StateGraph` as a node tomorrow with its middleware intact — no rewrite at the boundary.

**LangSmith is the other half.** The OSS is how you build the agent; LangSmith is how you find out whether it's any good. Framework-agnostic — it speaks OpenTelemetry.

RESULTS

# Every number labeled

Measured on this build · Modeled from assumptions · Unavailable until real traffic. The third category is the one worth reading.

| METRIC                                                                                                             | VALUE     | BASIS       |
|--------------------------------------------------------------------------------------------------------------------|-----------|-------------|
| Task resolution, benchmark<br>30 examples · 6 slices · every slice clean at this commit                            | 30 / 30   | MEASURED    |
| Cross-tenant data access<br>code evaluator, never a judge · hard gate                                              | 0 (30/30) | MEASURED    |
| Unapproved writes<br>no refund ticket without a human click                                                        | 0 (30/30) | MEASURED    |
| Escalation decisions correct<br>unsafe-resolution rate 0 · 4 labeled cases, so a percentage would be dishonest     | 4 / 4     | MEASURED    |
| Full-turn latency, p50<br>p95 7.1s — directional only at n=30                                                      | 3.8 s     | MEASURED    |
| Cost per priced eval turn<br>29 of 30 examples are single-turn · a real conversation costs several times more      | \$0.0010  | MEASURED    |
| Total API spend, whole build<br>OpenAI dashboard, not the estimate                                                 | \$ __. __ | MEASURED    |
| Deterministic tests passing<br>the security claims each have one behind them                                       | 139       | MEASURED    |
| Deflected labor / month<br>1,200 contacts · 60% billing-class · 70% containment · 6 min at \$0.60/min              | ~\$1,800  | MODELED     |
| Containment rate · recommendation attach rate<br>the North Star and the upsell line — both need production traffic | not known | UNAVAILABLE |

THE EXPERIMENT

Both arms resolved 30/30. I shipped flat.

Bars were pre-registered before the supervisor existed. It came in +2.8s on p50 against a +2.0s ceiling, with no mixed-intent gain and no quality gain to pay for it.

Last week this said 30 against 29, in the supervisor's favour, and I shipped flat anyway — one example on thirty isn't a quality difference. Rerun at this commit, they tie. That is what a one-example margin was worth. What reproduced both times is the overhead of the extra hop: 1.7x median latency, 61 tool calls against 45. The verdict rests on that.

PROVENANCE

|                   |                 |
|-------------------|-----------------|
| Commit, both arms | a29a530 · clean |
| Agent model       | gpt-5.6-luna    |
| Dataset digest    | 6ba144216f8d    |
| Evaluator digest  | decda2664959    |

Generated into reports/experiment.md, not typed. The script refuses to publish two arms that disagree on commit, model, dataset, or evaluators.